

Software Architecture and Design Coursework

GURKHAS MEMBERS:

VIJOHNSON

ERIK

SABRINA

THAKUR

Sprint 2 Report – Share Price Comparison System

1. Introduction

This project aims to design and develop a Share Price Comparison System that allows users to retrieve, store, and visualise stock price data over a selected time period. The system interacts with an external data source and supports local storage to enable offline functionality.

Sprint 2 focuses on deriving a software architecture from requirements and implementing the system using Clean Architecture principles.

2. Requirements Summary

- Retrieve share price data
- Store data locally
- Load stored data
- Display chart
- Compare companies (design level)

3. Business Concept Model

Entities include User, Company, SharePrice, DateRange, Chart, StoredData.

4. Use Case Model

Search Share Prices:

User enters symbol and dates → system validates → retrieves data → displays results.

5. System Interfaces

IShareRepository: getPrices()

IStorage: save(), load()

IChart: display()

IYahooAPI: fetch()

6. Business Type Model

Defines attributes for Company, SharePrice, DateRange, Chart, StoredData.

7. Initial Architecture

UI → Controller → Use Cases → Repository → API

Storage connected to Use Cases.

8. Interface Allocation

ShareRepository → IShareRepository

LocalStorage → IStorage

ConsoleChart → IChart

YahooFinanceAPI → IYahooAPI

9. Sequence Description

UI → Controller → Use Case → Repository → API → back → Save → Chart display.

10. Design Justification

Clean Architecture ensures:

- Separation of concerns
- Loose coupling
- Scalability
- Testability

11. Patterns Used

- Clean Architecture
- Repository Pattern
- MVC
- Adapter Pattern

12. Conclusion

The system is modular, scalable, and ready for Sprint 3 enhancements.